THE FALKLAND ISLANDS: NEW PALAEOMAGNETIC DATA AND THEIR ORIGIN AS A DISPLACED TERRANE FROM SOUTHERN AFRICA

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Abstract. Palaeomagnetic data from thirteen dolerite dykes of Lower Jurassic age from the Falkland Islands (52°S 300°E) yield a mean palaeomagnetic pole at 4.4°S, 272°E (a95 = 7.5°, calculated with N = 13 dykes, sampled at 29 sites). This pole position is far removed from the Mesozoic Apparent Polar Wander Path for South America. We interpret this in terms of relative displacement between the Falkland Islands and South America, with the islands formerly located to the east of the Transkei coast of southern Africa. This location is some 500 km northward and rotated anticlockwise 100° relative to the Cretaceous position for the islands inferred from simple closure of the South Atlantic. This inferred movement of the terrane containing the Falkland Islands would suggest that the Falkland Plateau is composed of several crustal blocks such that the block containing the islands was allowed to rotate independently of others. This clockwise rotation and southward translation of the Falkland Islands terrane was caused by the separation of the Antarctic and Africa plates, the early stages of this process being dominated by strike-slip motion between the two plates, between about 200 and 125 Ma years ago. This reconstructed position for the islands was originally suggested by Adie [1952] on the basis of its near perfect matching of the stratigraphy, structural geology and outcrop pattern of the eastern Karoo basin with that of the Falklands.

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Introduction

The Falkland Plateau is an extremely unusual morphological feature projecting out from South America at near right angles for over 1500 km, and bounded both north and south by steep escarpments. The geology of the plateau is mainly inferred from geophysical studies, a limited number of boreholes and from the exposed geology of the Falkland Islands. In Gondwanaland reconstructions this area has either been disregarded entirely or regarded as a contiguous part of South America; the latter assumption has led to problems of overlap with parts of West Antarctica and the Falkland Plateau when Gondwanaland reconstructions have been made [e.g. Norton and Sclater, 1979].

The Falkland Islands lie some 600 km east of South America and occupy a position slightly west of centre in the Falkland Plateau. The geology of the islands (Fig. 1) is composed largely of sedimentary formations ranging from Lower Devonian to Upper Permian (or Triassic) in age, with the exception of a basement gneiss complex exposed at Cape Meredith in the south of West Falkland which has yielded K-Ar ages of 950-980 Ma [Rex and Tanner, 1982]. The sedimentary formations have been subjected to a major episode of folding which has produced a dominant east-west axial fold trend in East

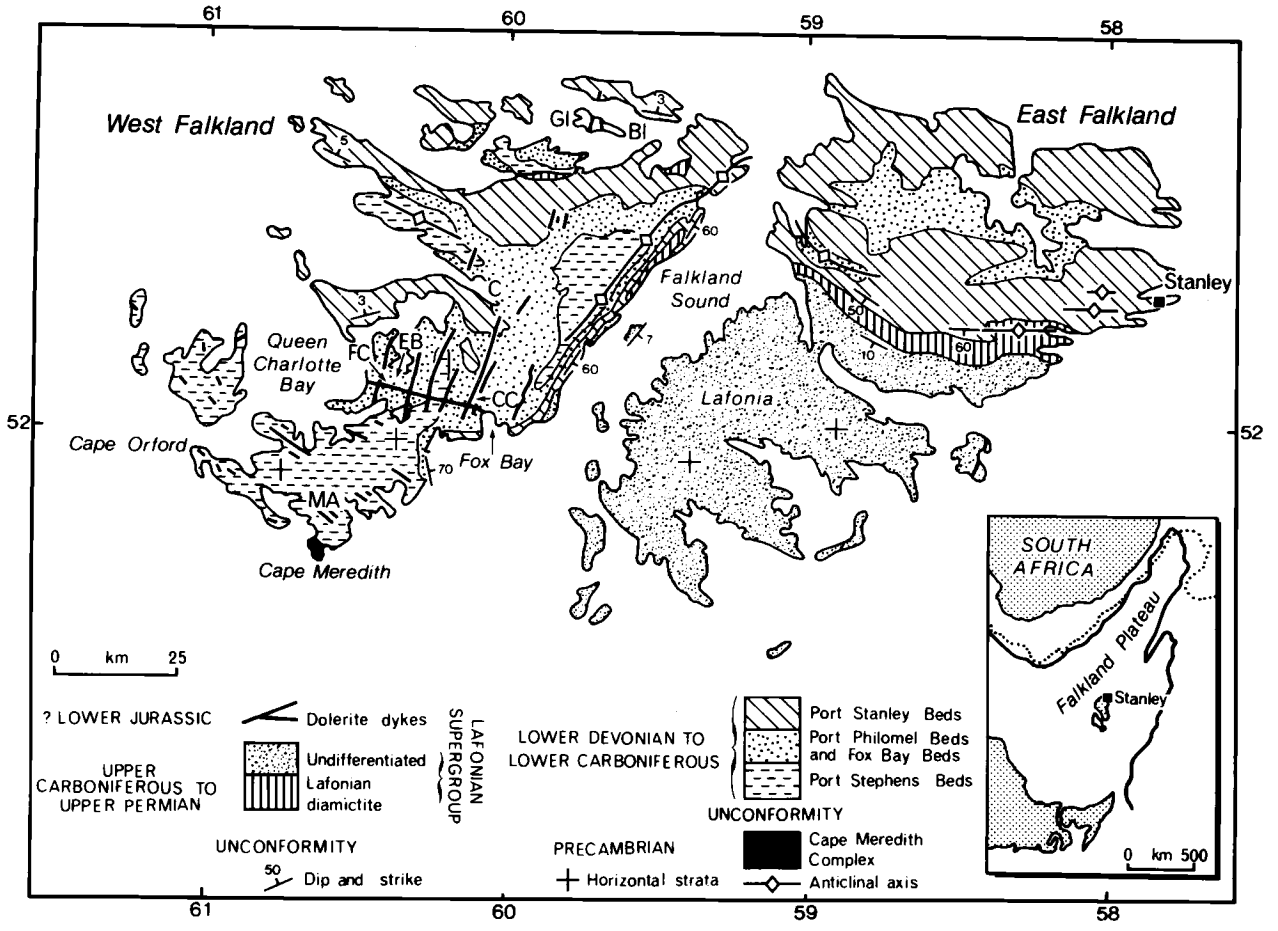


Fig. 1. Geological map of the Falkland Islands after Greenway 1972. Localities referred to are as follows: FC Fish Creek, EB East Bay, MA Mount Alice, CC Centre Camp, C Chartres, BI Broken Island, GI Golding Island. Inset, position of the Falkland Islands and Falkland Plateau prior to opening of the South Atlantic in Lower Cretaceous times (Based on Martin et al.,1981; 1982).

Falkland which swings around to north-south in parts of West Falkland [Adie, 1952; Greenway, 1972].

These rocks have been intruded at a later stage by two distinct dyke swarms, which trend approximately N-S and E-W, and are almost all confined to outcrop on West Falkland. The N-S swarm mainly occurs in central and northern parts of the island while the E-W swarm outcrops mainly in the south. Dykes from both swarms can vary from a few to tens of kilometres in length and from three up to a maximum of several tens of metres in width. They have straight outcrop patterns, regardless of the varying attitude and age of the country rocks they intrude, including cross cutting the fold axes of the country rocks, and are indifferent to topographical changes. This indicates that they post-date all other major tectonic events in the area and would tend to suggest that they have maintained a vertical attitude since their time of emplacement [Greenway, 1972]. These two dyke swarms show distinct differences in petrography; the N-S dykes are hypersthene-bearing and the E-W are olivine-bearing. Chemical analyses indicate that these swarms have distinctive geochemical signatures [Mitchell pers. comm., 1987]. The relative ages of these two swarms is uncertain, neither fieldwork nor photo-interpretation [Greenway, 1972] has been able to establish cross-cutting relationships. A single K-Ar determination on a dyke from near Cape

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Meredith has yielded an age of 192±10 Ma [Cingolani and Varela, 1976]. Neither set of dykes is well exposed inland, even though they are typically 10 to 15 metres wide and form ridges 2 to 3 metres above the surrounding land. Coastal exposure is usually good but rarely found despite the long coastline in the west-central part of West Falkland where such exposures would be expected.

A third distinct dyke swarm was identified during the fieldwork associated with this study. This swarm occurs in the southwest of West Falkland near Cape Orford (Fig.1) and on the adjacent islands; it is referred to here as the "Cape Orford" dyke swarm. These dykes, although relatively common, are virtually unexposed inland and are only seen on the coast where they are typically weathered out to leave a distinctive straight sided slot in the baked sandstones of the Port Stephens Beds into which they are intruded. In contrast with the N-S and E-W dyke swarms, the Cape Orford swarm shows no distinct preferred orientation, the dykes are narrow, usually less than 1 m wide, and are relatively badly eroded. Where they do cropout it is usually only the chilled margins that are preserved as upstanding sheets. Geochemical analysis indicates that they are quite distinct from the other dykes and are more intermediate in composition [Mitchell pers. comm., 1987]. They would appear to be limited in their occurrence to the outcrop of the Port Stephens beds of Devonian or older age in the south west of the island. This may indicate that they are substantially older than the N-S and E-W swarms.

Sampling and Laboratory Demagnetization Studies

The initial palaeomagnetic study of the Falkland dykes [Mitchell et al., 1986] was based on 10 hand samples from 4 N-S trending dykes and 4 samples from the baked margins of a single E-W dyke. The latter samples had poor stability to laboratory demagnetization and yielded poorly defined, widely dispersed and negatively inclined directions. The N-S dykes however yielded consistent results which gave a pole position anomalous with respect to the South American Apparent Polar Wander (APW) path [Mitchell et al., 1986]. The intention of this study was to confirm or refute this result from the N-S dykes, such that any possibility that the original result was a local anomaly from one part of the island or that misorientation of the samples occurred during collection could be entirely eliminated. This was to be achieved by sampling more dykes at a larger number of localities over a wider geographic area using both sun and magnetic orientation of core and hand samples. The difficulties of transporting the drilling equipment in the field combined with the sparsity of outcrop imposed severe restrictions on the number of samples actually recovered. A total of 239 oriented drill cores were collected from 30 sites in 18 individual dykes (5 N-S, 3 E-W and 10 in the Cape Orford swarm) and an additional two N-S dykes were again hand sampled. Subsequently an independent collection of 23 hand samples from 21 dykes (9 N-S, 2 E-W, and 10 Cape Orford) was made by P.J. Rickwood.

The majority (c. 80 - 85%) of samples have been measured using either stepwise thermal or A.F. demagnetization using a re-entrant cavity type SQUID magnetometer [Shaw et al., 1984]. The remaining samples were treated by continuous thermal (measure while heating) demagnetization using a biomedical probe type SQUID magnetometer [Share, 1987], initially to compare the validity of the results with those from conventional thermal demagnetization and then to acquire thermal demagnetization data more rapidly. All directional results were analysed by means of stereonet and orthogonal projections combined interactively with Principal Component Analysis on a microcomputer. All samples have been treated individually, the normal A.F. demagnetization interval being 5 mT up to 100 mT and 50°C steps up to 500°C in thermal demagnetization, beyond which smaller steps were used up to a maximum of 700°C. In continuous thermal demagnetization experiments the normal measurement interval was 20-30°C up to a maximum of 720°C. Individual specimen directions were regarded with caution if less than five demagnetization steps contributed to the resolution of a characteristic component direction, indeed only low temperature/low coercivity directions have been determined with less than this number of demagnetization steps. We are therefore confident that we have identified and eliminated any spurious magnetizations from our final analysis.

Results From the North - South Dyke Swarm

Detailed sampling of three N-S dykes was undertaken in the East Bay area (Fig.

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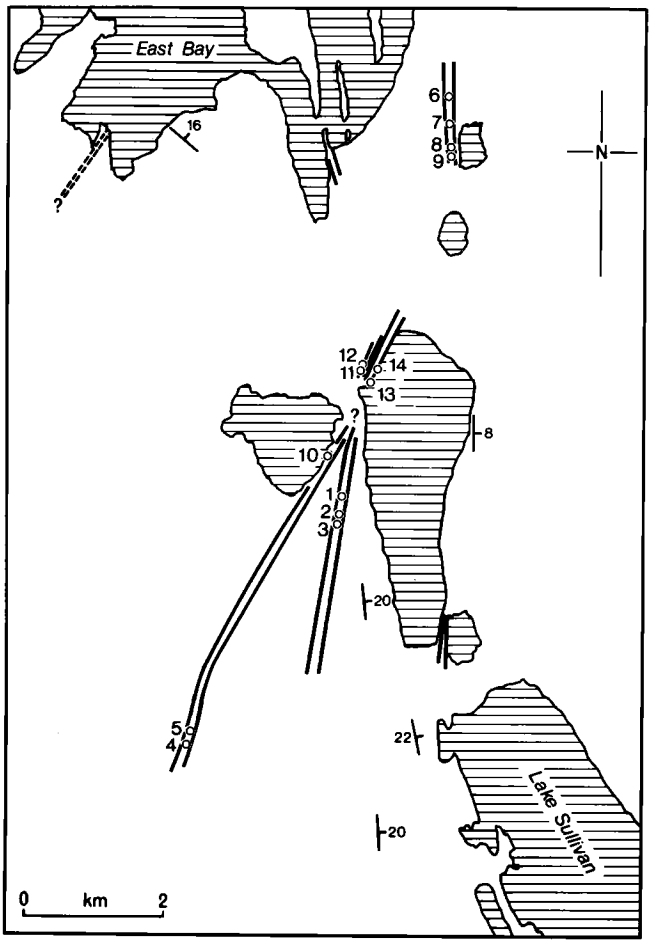


Fig. 2. Geological sketch map of the area between East Bay and Lake Sullivan, showing the palaeomagnetic sampling sites (Table 1) and the inferred and exposed dykes of the area. The country rocks are sandstones with minor shales of the Fox Bay Beds.

2) in order to investigate within-site and between site variation from the same and nearby dykes. The nature of the exposures was such that no samples of baked margin metasediments could be recovered to perform a baked contact test as was intended. The samples show a wide range of NRM intensities, although their stability to demagnetization was found to be independent of total NRM intensity. Linear demagnetization trajectories were recovered using all three demagnetization techniques employed.

As defined by demagnetization steps at low temperatures or low alternating fields, secondary magnetic components are poorly clustered about the present day field or otherwise randomly directed upwards, probably reflecting acquisition of minor viscous components during storage in the laboratory prior to demagnetization (Fig. 3). Median destruction fields are all less than 40 mT, with low coercive components typically being removed at less than 25 mT (e.g. sample EB14.2, Fig. 4). Comparison of sister specimens or samples treated by stepwise and continuous thermal demagnetization shows very consistent behaviour (e.g. specimens EB12.4 and EB12.04, Fig. 4). The characteristic direction is usually recovered between 200°C and 550°C, and is consistent with the response to A.F. demagnetization treatment. Thermomagnetic analyses yield Curie points of 550°C to 580°C which, together with the thermal and A.F. behaviour, are consistent with magnetite being the main magnetic carrier.

A north-westerly and moderately to steeply inclined downward direction was recovered from all sites with the exception of sites EB10 and EB13 which show mixed polarity with several samples containing opposed directions (Table 1). The mean direction for this group of dykes based on a site unit weighting analysis yields D = 325.5°, I = 57.3° and a95 = 6.6°, n = 12 sites. This result is

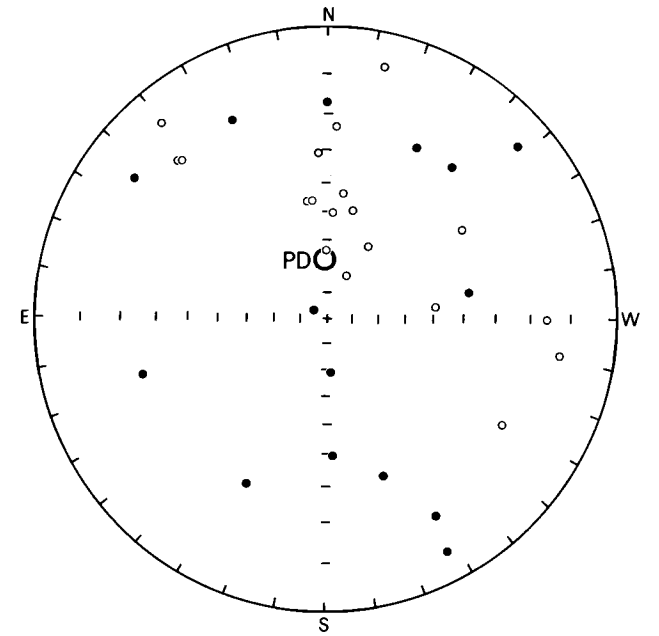


Fig. 3. Equal area stereonet projection of the isolated low temperature or coercivity components from the East Bay collection. Note the clustering about the present geocentric axial dipole day field (PD) direction. Closed (open) symbols are on the lower (upper) hemisphere.

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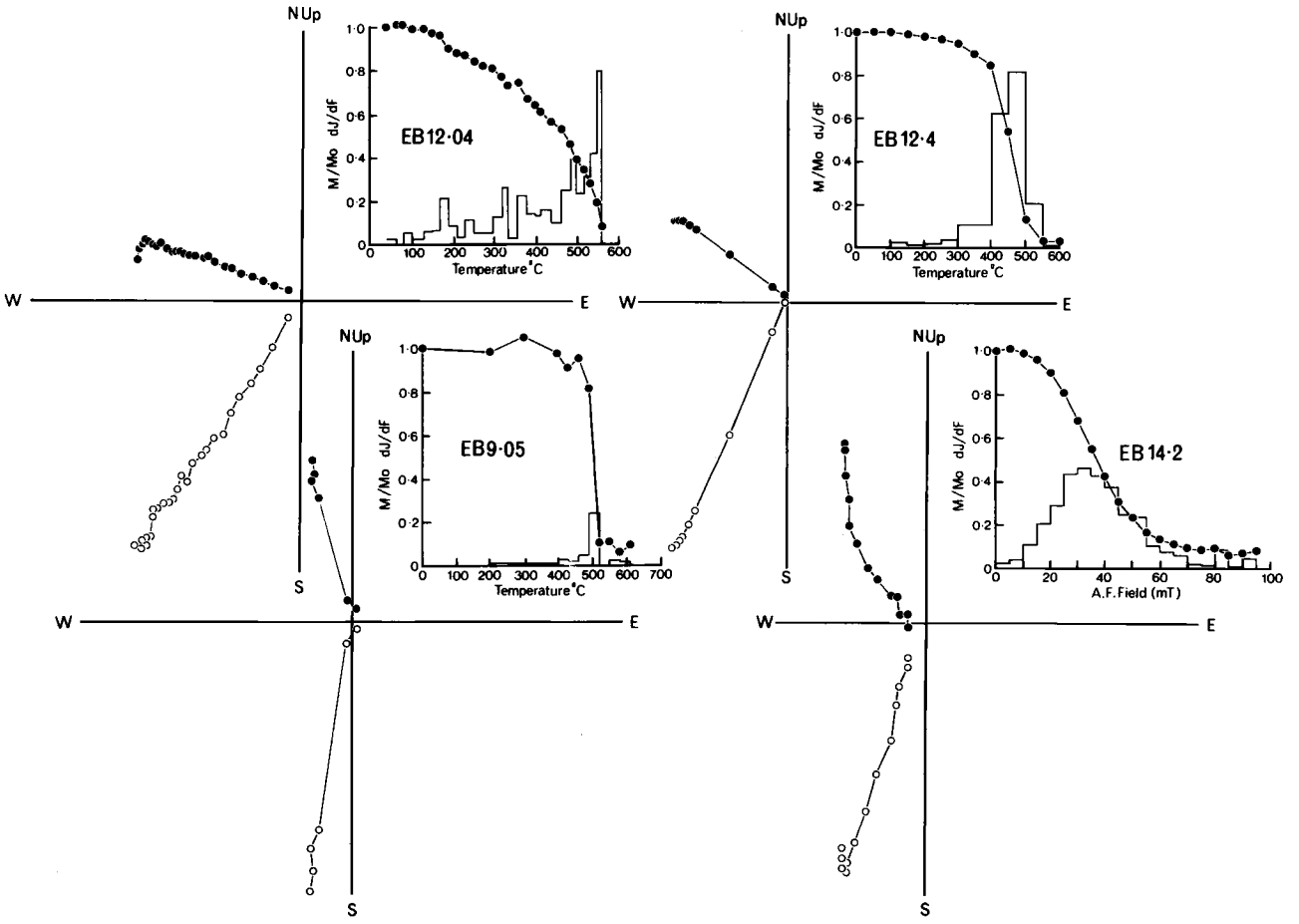


Fig. 4. Orthogonal projections of demagnetization data for samples from the East Bay area. Open (closed) symbols represent the projection of the vector endpoints on the vertical vs. east-west (north-south vs. east-west) plane. Also shown are normalised intensity decay curves, crosses represent data shown on the orthogonal diagrams, pluses data not included, histograms represent simple unblocking spectra. Samples EB9.05 and EB12.4 show typical responses to thermal demagnetization, while EB14.2 illustrates the A.F. demagnetization response. Sample EB 12.04 is a sister specimen to EB12.4 but demagnetized using continuous thermal demagnetization and measured at 20°C intervals, note how well they compare.

similar to our previous results from the Fish creek area (D = 342.8°, I = 61.1°, a95 = 9.6°, n = 9 independent block samples from four dykes; Mitchell et al., 1986] and to the limited work we have done on the new collection by P.J. Rickwood (D = 319.4°, I = 56.8°, a95 = 11.8°, n = 11 block samples from seven dykes). A site by site analysis yields a result with higher precision, essentially due to the larger number of sites than dykes involved, but gives undue weight to the dykes sampled at East Bay because of the high number of sites per dyke collected in this area. Combining all the results, giving individual dykes unit weighting (Table 2 and Fig. 7A), yields a mean direction of D = 324.1°, I = 56.5°, a95 = 7.5° with n = 13 dykes, resulting in a palaeomagnetic pole at 4.4°S and 272.0°E with dp, dm = 7.8°, 10.8° respectively.

Results From the East - West and Cape Orford Dyke Swarms

The east-west swarm was sampled in the vicinity of Mount Alice in southeastern West Falkland (Fig. 1). Due to the extremely limited exposure only four

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TABLE 1. Results from EAST BAY -- North-South Dyke Swarm

SITE	N	n	R	DEC	INC	k	a95	dyke
EB1		4	3.94	316.5	53.9	51.5	12.9	1
EB2		2	1.99	302.7	49.8			1
EB12		4	3.97	306.8	61.9	102.9	9.1	1
EB4		6	5.79	334.4	45.7	23.4	14.1	2
EB5		4	3.80	326.4	59.8	15.4	24.2	2
EB10		4	3.92	344.9	40.3	36.2	15.5	2
EB13		6	5.97	319.1	71.6	152.5	5.4	2
EB14		4	3.90	322.9	46.6	30.7	16.9	2
EB6		3	2.99	329.0	61.6	139.1	10.5	3
EB7		5	4.91	317.7	65.2	45.8	11.4	3
EB8		6	5.85	331.3	67.4	33.9	11.7	3
EB9		6	5.74	343.9	55.2	19.4	15.6	3
MEAN	12		11.79	325.5	57.3	44.2	6.6	
Published result from N-S dykes at Fish Creek House								
MEAN		9	8.73	342.8	61.1	29.5	9.6	
N-S dykes collected by P.J. Rickwood 1987								
MEAN		11	10.37	319.4	56.8	15.89	11.8	

Individual sites given unit weighting for comparison of means from the thre separate collections. N is the number of individual collecting sites, n the n samples measured; R the length of the resultant vector; Dec, Inc are the mean declination and inclination respectively; k Fisher's precision parameter; a95 radius of the 95% confidence cone about the mean direction. Dyke is the arbri reference number assigned to each dyke. Note that sites EB10 and EB13 contain directions that have been inverted for compatibility.

TABLE 2. Analysis of the North-South dyke swarm giving unit weighting to each dyke.

DYKE		N	R	DEC	INC	k	a95	VGP	
Fish Creek	A	4	3.90	350.2	64.5	29.6	17.2	-8.7	293.2
Fish Creek	B	3	2.95	331.2	65.0	41.6	19.6	-12.0	280.4
Fish Creek	C	1		336.3	39.0			13.1	277.5
Fish Creek	D	1		159.5	-24.5			22.8	278.3
East Bay	1	3	2.98	309.9	55.8	100.1	12.4	-8.6	261.3
East Bay	2	4	3.98	331.5	62.7	120.4	8.4	-9.2	279.7
East Bay	3	5	4.88	331.4	53.1	33.0	13.5	0.8	276.5
Centre Camp		1		335.3	56.3			-1.4	280.5
Golding Is.		2		336.5	51.1			4.0	280.1
Broken Is.		1		310.8	42.3			2.5	256.4
L. Hog Ground		1		298.6	47.9			-7.1	249.3
Teal River		1		336.5	51.1			3.7	280.1
Gun Hillside		1		328.6	78.9			-32.8	286.9
Cape Meredith		1		298.2	44.6			-5.0	247.5
Mean		13	12.62	324.1	56.5	31.8	7.5	-4.4	272.0

Note that Fish Creek dyke D has been excluded from the analysis and the result from Cape Meredith inverted. N is number of sampling localities (or hand specimens in those dykes collected by that method). VGP is the latitude (^{O}N) and longitude (^{O}E) of the Virtual Geomagnetic Pole. Other notation as in Table 1.

sites could be sampled from three dykes. Their demagnetization behaviour is more complex than seen in the N-S swarm, in that most illustrate complex multicomponent demagnetization histories, particularly when treated using continuous thermal demagnetization. The most common behaviour consists of a rapid loss in intensity below 250°C, a gain in intensity above 500°C followed by another decrease by 680°C (e.g. samples MA1.2 and MA2.2 in Fig. 5). In most samples three components can be resolved, normally in the temperature ranges of 20°-300°C, 250°-540°C and 560°-680°C.

Analysis of the resolved component directions shows a coherence of both high and low temperature components in sites MA1 and MA3 (from the same dyke but separated by some 2 km). Although well defined in individual samples the intermediate component is inconsistent in three sites, and only poorly grouped in site MA1 (Table 3 and Fig. 7B). The low temperature component in site MA2 has a similar direction to those isolated in sites MA1 and MA3, while the high temperature components in site MA2 and MA4 are also comparable but markedly different in direction from the other two sites. Thermomagnetic curves (Fig. 6) and demagnetization behaviour would seem to indicate that hematite is responsible for the high temperature component while magnetite may carry the low and intermediate temperature components. It is worth noting that in the previous study, samples from the baked margins of an E-W dyke also produced negatively inclined directions but were inconsistent with each other. The lack of consistency in results from the different east-west dykes is difficult to explain without further detailed magnetic experiments and sampling.

Of the ten dykes/sites sampled in the Cape Orford swarm three consist entirely of baked sediments from the margins of unexposed dykes. These samples have all proved too weak to measure or unstable to conventional stepwise A.F. and thermal demagnetization. Of the remaining seven sites only four have so far yielded well defined results (Table 3 and Fig. 7C). The mean direction is D = 235.2°,I = 42.9°, a95 = 13.6°, n = 4 dykes/sites. This component was typically isolated in the temperature range 200°C-580°C or between 20 mT and 80 mT during A.F. demagnetization, suggesting magnetite to be the principal magnetic carrier present. This direction is markedly different from that recovered from the N-S and E-W dyke swarms. This result is consistent with the geological and geochemical observations that suggest that the Cape Orford dyke swarm is distinct from the other dyke swarms.

Interpretation of Results

The additional results from the N-S dyke swarm confirm the northwesterly positively inclined direction previously obtained in the preliminary study [Mitchell et al. 1986]. One problem with minor intrusive units of this type is the uncertainty over their rate of cooling and consequently has the sampling scheme averaged out palaeosecular variation (PSV) adequately? Although no recent model exists for PSV in Jurassic times, comparison with Cretaceous and Tertiary data, for a latitude of 37° (the palaeolatitude calculated from the mean inclination) would suggest PSV to be of the order of 14° to 19° [Merrill and McElhinny, 1983]. PSV from a data set such as this can be estimated by calculating the angular standard deviation of the virtual Geomagnetic Poles (VGP) of the individual units. The value of 18.8° derived from this data set would therefore tend to suggest that we have successfully accounted for PSV when calculating the mean direction. In addition, the dual polarity results from sites EB10 and EB13, the normal polarity of the dyke from Cape Meredith, the consistency of the results from a geographically widespread area and from a reasonable number of dykes, would all suggest that we have obtained a representative sample of the palaeomagnetic field direction.

The single K-Ar date from the dykes indicates that they were emplaced in Lower Jurassic times. This age is consistent with the widespread emplacement of basic intrusive bodies throughout southern Gondwanaland at this time [Schmidt and Rowley, 1986]. The pole position yielded by the dykes does not however lie on the Mesozoic APW path for South America [Fig. 8; Vilas, 1981; Irving and Irving, 1982] and we therefore appeal to a tectonic solution to this problem.

We previously suggested [Mitchell et al.,1986] a position for the islands to the east of the Karoo basin in southern Africa (Fig. 9). This position is arrived at after rotation of the islands as a contiguous part of South America, such that they occupy a position to the south of Africa [Martin et al., 1982], followed

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THE FALKLANDS, NEW PALAEOMAGNETIC RESULTS

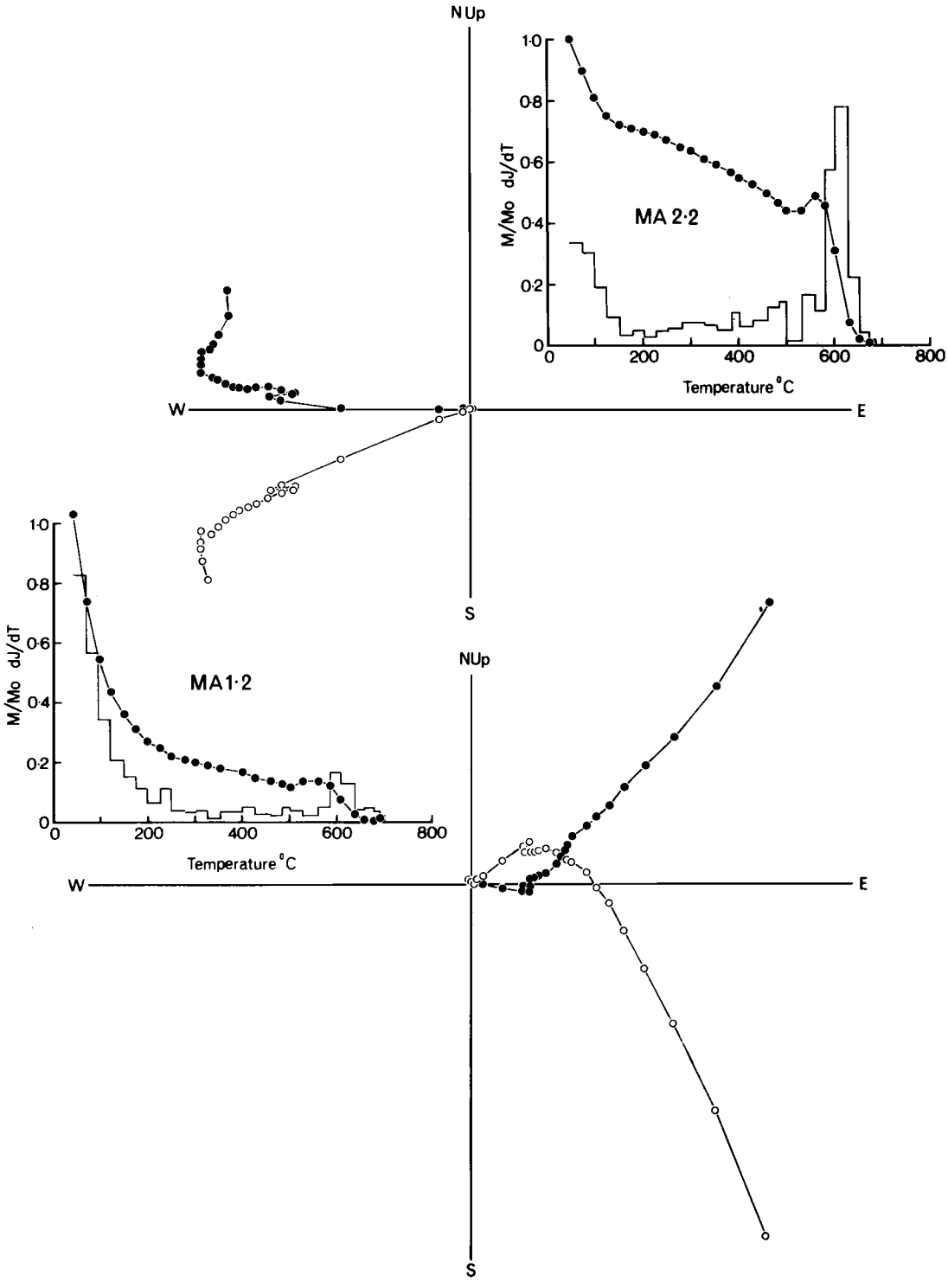


Fig. 5. Demagnetization diagrams for samples from the E-W dyke swarm at Cape Orford. Note the rapid loss in intensity below 250°C and the peak at approx. 600°C. Conventions as in figure 4.

by a 100° anticlockwise rotation about a pole within the block and northward translation of some 500 km. Recalculating the mean direction (Dec 166°, Inc 56.5°) then yields a north palaeomagnetic pole at 280°E 78°N. This is comparable to the results obtained from Triassic to Jurassic rocks from Africa (Table 4)

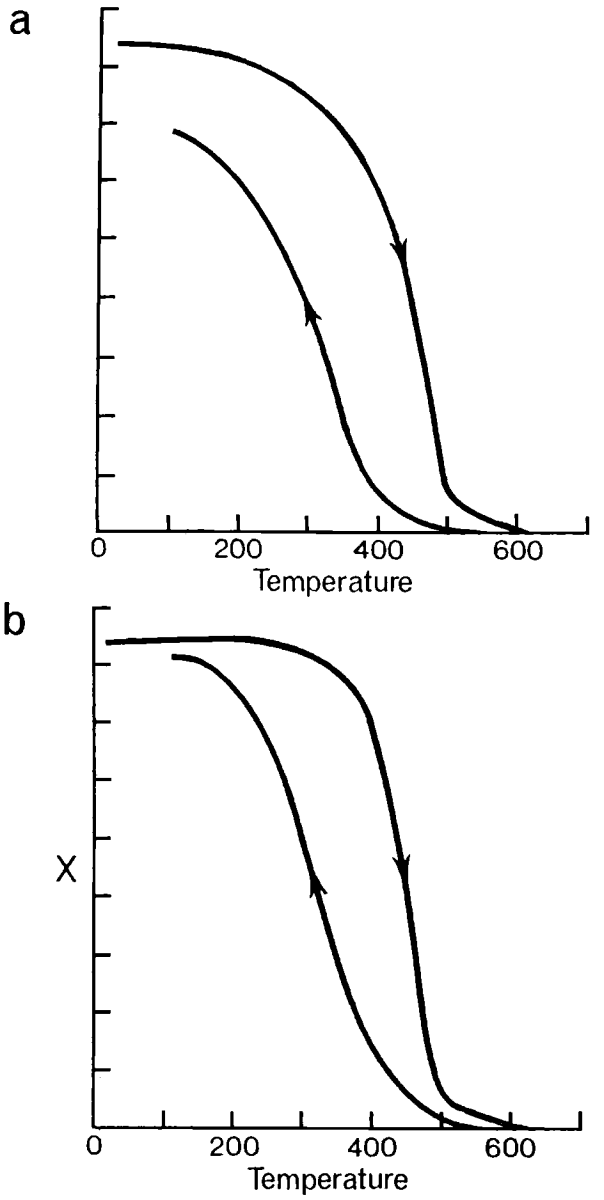


Fig. 6. Thermomagnetic curves for samples from the E-W dyke swarm. Note the presence of a slight high temperature "tail" indicating the presence of hematite in these samples.

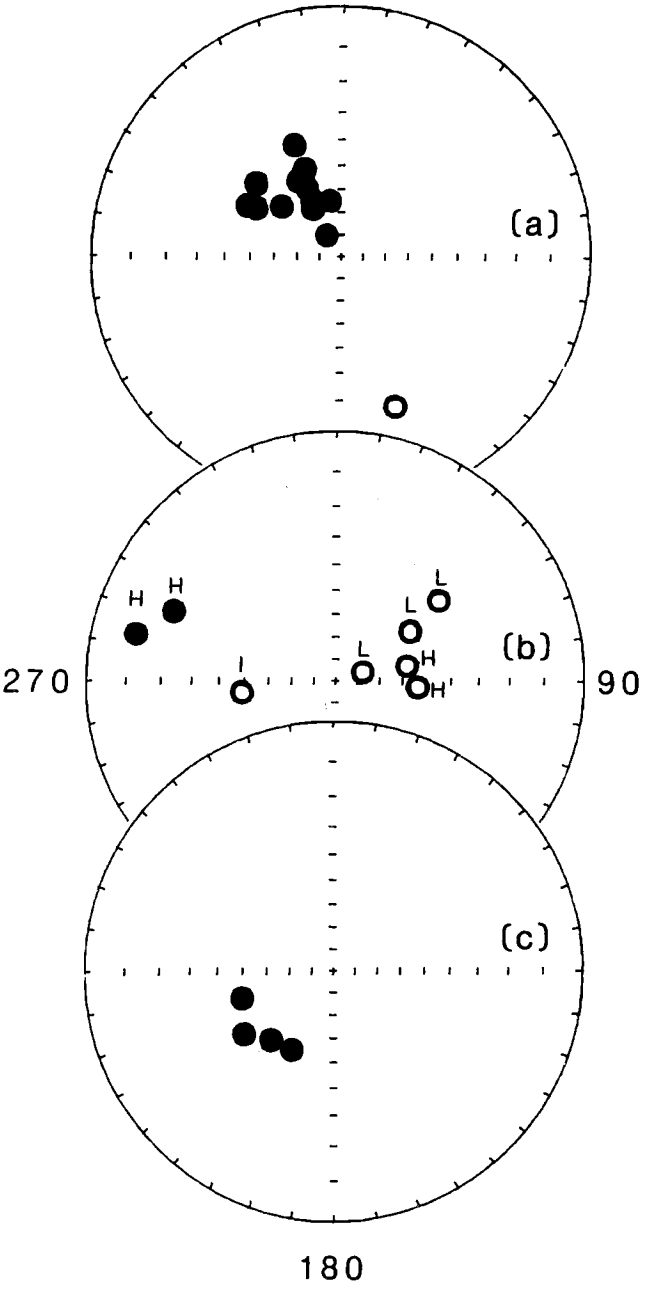


Fig. 7. Distribution of characteristic mean directions; a) is dyke mean directions for the North-South dyke swarm, b) low, intermediate and high unblocking temperature components in the East-West dykes from Mount Alice, and c) the mean directions from the four dykes yielding stable directions from the Cape Orford swarm.

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TABLE 3. Results from the East-West and Cape Orford swarms.

SITE	N	R	DEC	INC	k	a95	Comp
MOUNT ALICE -- East-West Dykes							
MA1	5	4.90	69.1	-77.4	38.4	12.5	<250°C
MA1	5	4.51	263.6	-49.0	8.2	28.5	200-500°C
MA1	7	6.62	93.4	-52.8	15.8	15.6	>550°C
MA3	6	5.55	56.9	-50.4	11.2	20.9	<250°C
MA3	7	8.84	78.5	-57.4	36.9	10.0	>250°C
MA2	6	4.94	52.6	-34.8	4.7	34.6	<250°C
MA2	7	6.87	293.5	19.5	45.6	9.0	>550°C
MA4	6	5.98	283.5	11.6	323.6	3.7	>550°C
CAPE ORFORD -- Dyke Swarm							
OR1	8	7.96	223.3	49.8	178.3	4.2	
OR2	7	6.71	235.2	42.9	20.8	13.6	
OR3	5	4.97	208.7	50.4	138.4	6.5	
OR4	6	5.59	253.4	48.1	12.2	19.9	
MEAN	4		235.2	42.9	20.8	13.6	

Notation as in Table 1. Comp refers to the direction isolated at the given unblocking temperatures.

TABLE 4. Jurassic - Triassic palaeomagnetic results from Africa.

	Palaeomagnetic North pole Lat(°N)	Long(°E)	AGE(My)
Hoachanas Lavas	61.9	251.9	161-173
Mateke Hills Complex	59	260	168
Karoo Dolerites	55.5	255.0	154-190
Sahara Dolerite Sills	72.0	261.0	160-180
Marangduzi Complex	70.2	285.1	182-196
Diabase Intrusives Liberia	68.5	242.4	173-192
North Mauritania Hank	69.4	232.0	187.0
South Mauritania Hodh	71.4	240.2	187.0
Draa Valley Sills, Morocco	65.5	230.5	180-186
Foam Zguid dykes, Morocco	58.0	259.0	182-187
Karoo Lavas	57	264	J1
Stormberg Lavas	71	269	J1-Tr
Freetown Igneous Complex	80.9	224.6	>180

After Brock (1981).

which themselves show a wide spread in pole positions. The discrepancy between the actual results from the Karoo and Falkland dolerite dykes is, we believe, due primarily to the fact that the Karoo result is based on a low number of samples, few of which were actually demagnetized.

This reconstructed position for the Falkland Islands terrane was originally suggested by R.J. Adie in 1952, who based his interpretation on four distinct lines of evidence. (1) Correlation of the stratigraphy and sedimentology of the Devonian to Triassic sequences of the Karoo basin with those exposed in the

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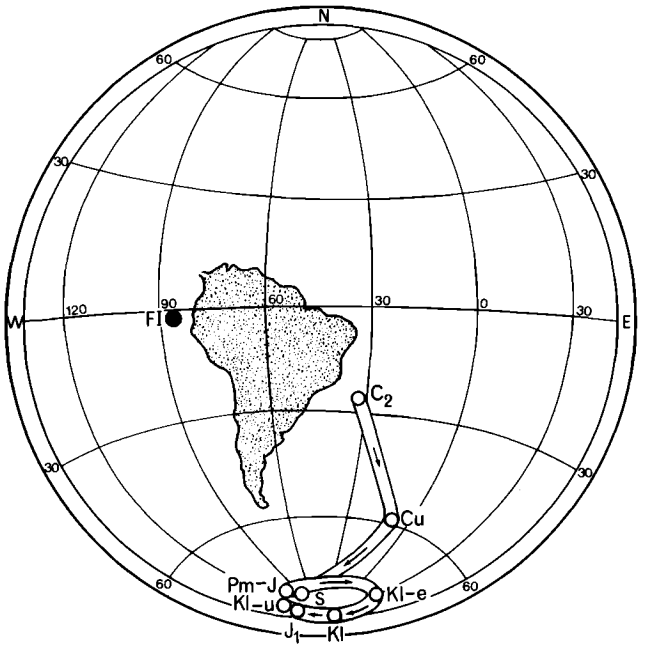


Fig. 8. The in situ pole position for the Falkland Islands does not coincide with the Apparent Polar Wander Path for South America. The solid path is that of Vilas [1981] (which is in agreement with that of Irving and Irving [1982] during this period).

Falklands. (2) The similar petrography and occurrence of dolerite dykes (two principal dyke swarms at right angles to each other) both in the Karoo and in the Falklands. (3) Correlation of the Falklands fold episode with those seen in the Cape Fold belt. (4) The continuity of ice transport directions in the Dwyka, southern Africa, and Lafonian, Falkland, diamictites after repositioning the Falklands to the east of the Transkei coast. Added support for these original criteria comes from a single K-Ar dating of the Falkland dolerites which gave a Karoo age [Cingolani and Varela, 1976] and a recent reinterpretation of the Permo-Carboniferous glaciation of southwestern Gondwana [Visser, 1987]. Visser's results indicate that a position for the islands to the east of the Transkei coast in southern Africa is consistent with the known glaciation data. In addition, this reconstruction juxtaposes the Cape Meredith gneiss complex, which has yielded ages of 950 - 980 Ma, with the Natal basement of south-east Africa that is of similar age [Rex and Tanner, 1982]. The Lafonian diamictite sequence (Fig. 1) contains plutonic and metamorphic clasts derived from a SW source terrane [Frakes and Crowell, 1969]. We also noted NE transport directions in the Port Stephens Beds which would suggest derivation from a pre-Devonian source to the SW. Such a source can be far more readily found in southern Africa after rotation than by maintaining the islands' present position with respect to South America.

Discussion

Reconstructions based on closure of the South Atlantic are constrained by a wealth of geophysical and seafloor geomorphological data [e.g. Martin et al.,1981; Lawver et al., 1985; Martin and Hartnady, 1986] and place the Falkland Plateau in a position immediately south and west of southern Africa (Fig. 1 inset). These reconstructions are essentially limited by the matching of Cretaceous marine anomalies or continental features, magnetic anomaly M10 (122Ma) being the oldest so far recognised and used [Martin and Hartnady, 1986]. Even in this reconstruction there is overlap between the Falkland and Agulhas Plateaus [Martin and Hartnady, 1986]. The most recent Jurassic palaeomagnetic data from the Ellsworth-Whitmore Mountains (EWM) crustal block of West Antarctica, also shows overlap between it and the Falkland Plateau. This overlap can only be minimised by moving the EWM as far south as possible within the constraints of the palaeomagnetic data [Grunow et al., 1987a]. The separation of the Falkland Plateau into several distinct terranes in Jurassic and early Cretaceous times would help to resolve some of these reconstruction problems.

The area of Southern Gondwanaland between South America, Africa and the Antarctic craton is regarded as having a complex history during the separation of the Antarctic and African plates and was the source area for a number of terranes now forming part of West Antarctica [e.g. Dalziel and Elliot, 1982; Grunow et al., 1987a,b]. It has experienced widespread igneous activity (e.g. the Karoo and Ferrar igneous suites) associated with rifting and strike-slip faulting in the Triassic and Jurassic. At least three major strike slip fault zones intersect in this area. These faults separate the African from the Antarctic plate, Africa from the Falkland terranes and the third, the Transantarctic Fault zone, divides East and West Antarctica. These faults would have bounded this area of initial

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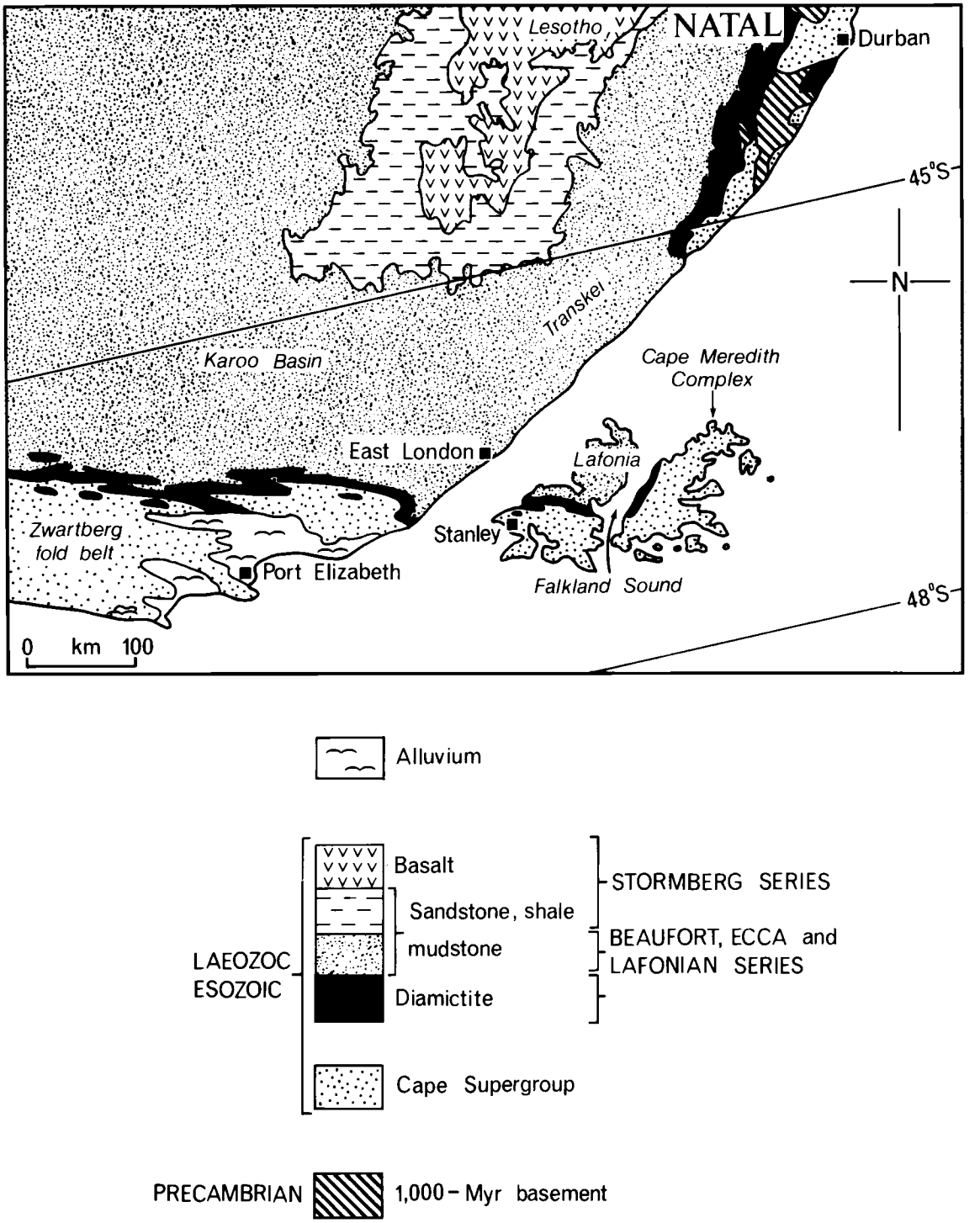


Fig. 9. Reconstruction of the position of the Falkland Islands relative to southern Africa in the Lower Jurassic. Note the exceptionally good continuity of geological units. Palaeolatitudes are based on McElhinny et al., 1968 and Brock 1981.

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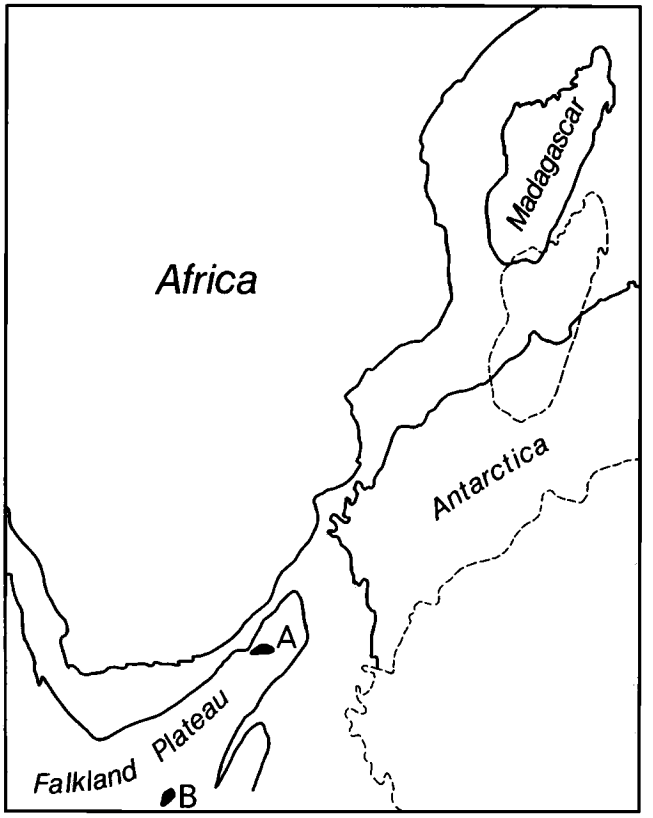


Fig. 10. Reconstruction of Southern Gondwanaland illustrating the dextral strike-slip motion between Antarctica and Africa during the Cretaceous. Africa is held fixed and the outline of the position of Antarctica at anomaly M21 (145 Ma - dashed) and M10 (122 Ma - solid) times is shown. [After Martin and Hartnady, 1986]. The reconstructed Jurassic position of the Falkland Islands is marked A and the Cretaceous position B.

rifting and extension, first between Africa and Antarctica and later between Africa and South America. Since the early movement between Antarctica and Africa was dominated by dextral strike-slip motion (Fig. 10) it is hardly surprising to find terranes that originally lay between these areas have suffered significant rotations.

At least two terranes that originated in this area are believed to be partially underlain by oceanic crust, these are the Agulhas Plateau [Allen and Tucholke, 1981] and the Falkland Trough [Ludwig, 1983]. We therefore propose that the Falkland Plateau is an amalgamation of terranes, which may in part be underlain by oceanic crust, and that the Falkland Islands microplate has suffered displacement and rotation from a position east of the Transkei coast in southern Africa as originally suggested by Adie [1952]. Translation of the Falkland Islands block would have occurred after the last dated fold episode in the Cape Fold belt of 215 Ma [Halbich et al., 1983] and/or the single K-Ar age for dyke emplacement 192 Ma but prior to 122 Ma years ago (M10 magnetic anomaly times). After this time no relative movement occurred between the Falkland Plateau and South America [Martin et al.,1982; Martin, 1984]. There is therefore a period of some 67 to 90 My during which translation of the Falkland Islands terrane was achieved. The tight constraints on the timing and translation of this displaced terrane are such that we believe it to be unique in the sense that it can be returned to a near perfect reconstruction prior to displacement.

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